Synthetic PET Image Generation Using GAN and cGAN for Data Augmentation

Lina Babayan
M1, Faculty of Informatics and Applied Mathematics, UFAR
UFAR · Université de Toulouse · IRIT
Supervisors: Amel Aissaoui, Kristina Sargsyan

Problematic

Growing use of deep learning in medical Imaging

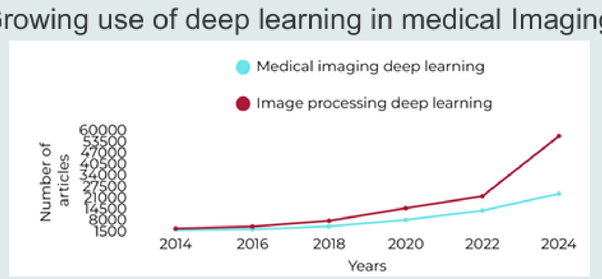
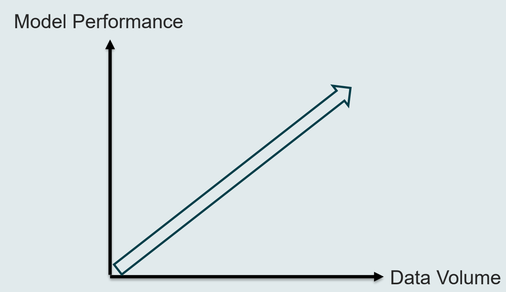


Figure 1: Growing use of deep learning in medical imaging. ScienceDirect.com | Science, health and medical journals, full text articles and books.

More training data improves model performance.



Limited PET[1] Data → Model limitations

Caused by:
- Expensive PET acquisition
- Privacy constraints
- Expert tumor annotation
- Rare pathological cases

- Overfitting
- Bias
- Poor generalization

[1] Positron emission tomography

Objective

Study synthetic PET image generation as a data augmentation strategy for limited PET datasets, while preserving anatomical and lesion-related consistency.

State of the Art

Traditional Data Augmentation

Rotation / flipping
Cropping / scaling
Intensity transformation

Limitations

visual realism ≠ tumor-structure consistency

Gap persists

GAN[2]-based Synthetic

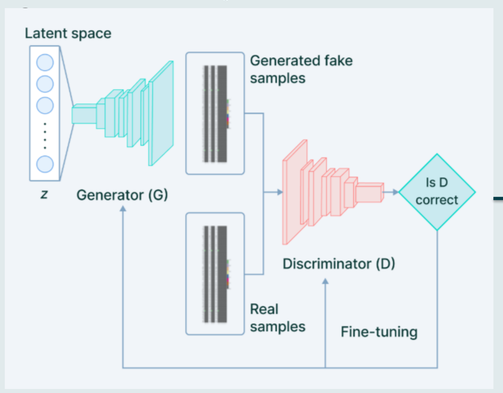


Figure 2: general GAN architecture
The Essential Guide to Neural Network Architectures

Related direction:

Conditional generation in medical imaging

[2] Generative adversarial network

Methodology

Four generation strategies were compared:
GAN baseline, then three conditions were compared:
mask only, CT[4] only, and CT + mask

Data: 33 patients → 676 valid 2D slices after mask filtering.

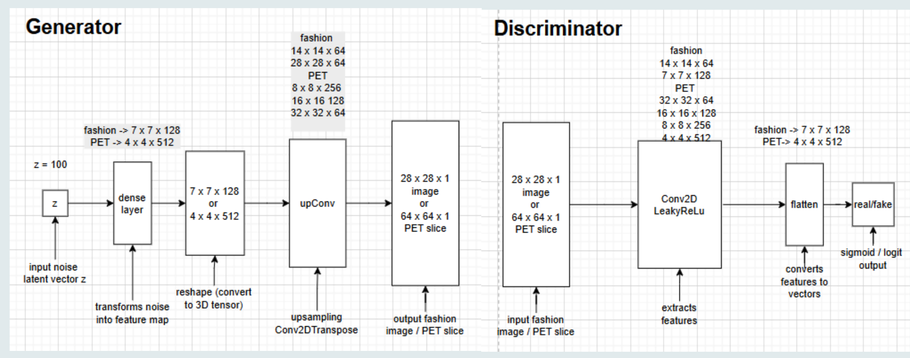


Figure 3: PET GAN architecture

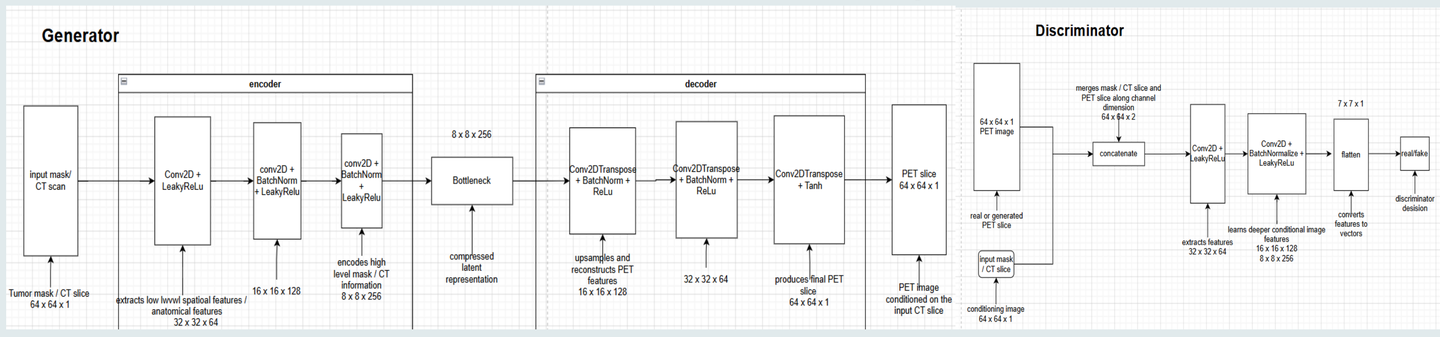


Figure 4: PET cGAN with mask-only / CT-only condition generator architecture

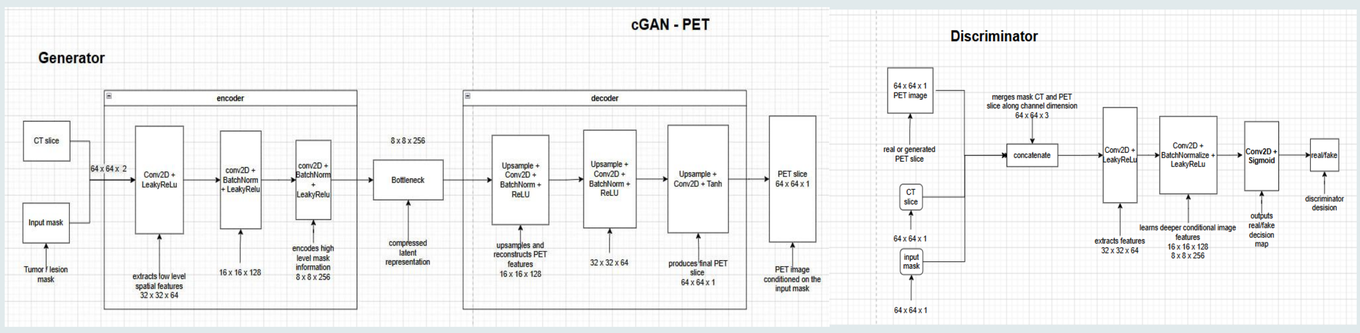


Figure 5: PET cGAN with mask and CT condition architecture

[3] Conditional GAN
[4] Computed tomography

Results

GAN generation

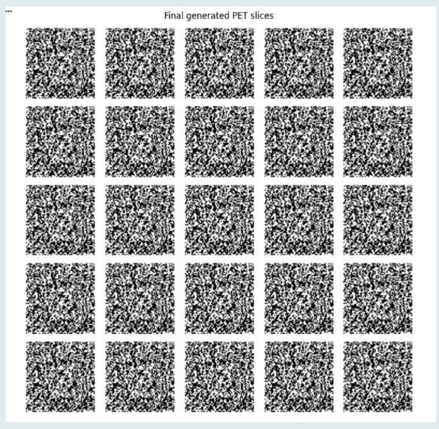


Figure 6: PET GAN output

cGAN generation with CT

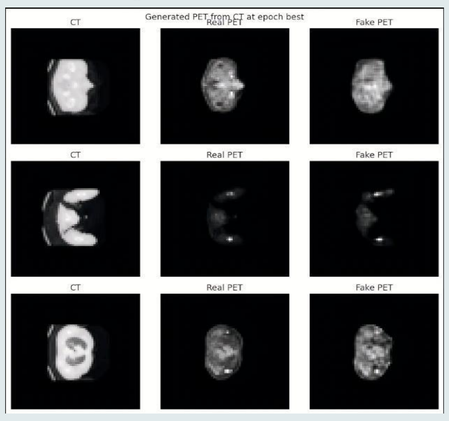


Figure 8: PET cGAN with CT-only conditioning

cGAN generation with mask

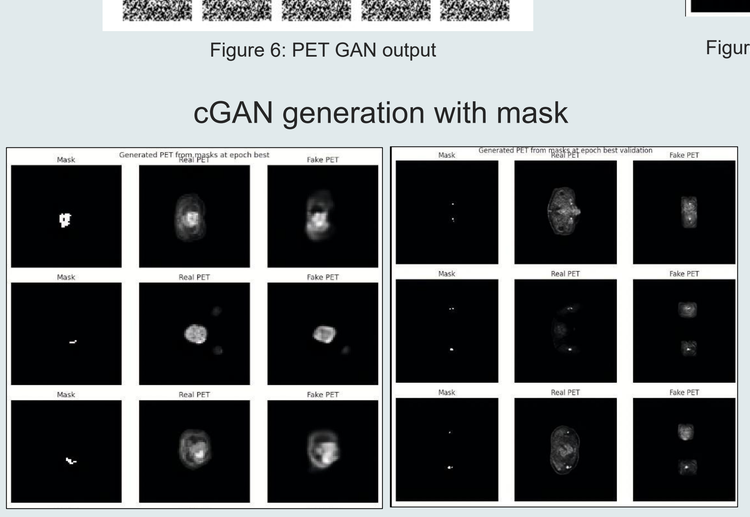


Figure 7: PET cGAN with mask-only conditioning - 2 cases

cGAN generation with CT and mask

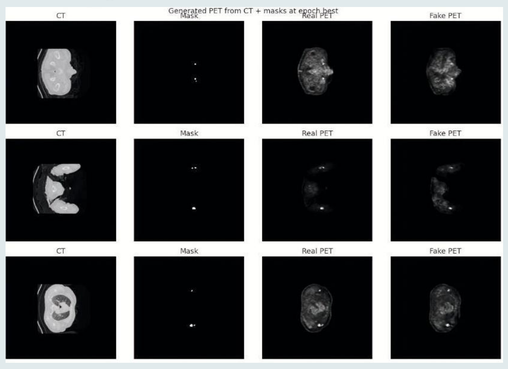


Figure 9: PET CT + mask cGAN results

Image resize ratio and minimum mask pixels experiments outcomes

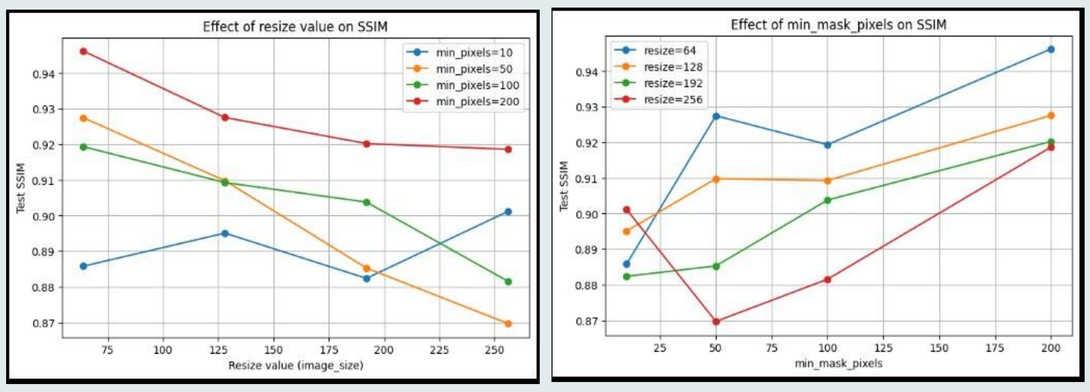


Figure 13: Experiment results(SSIM)

CT-only achieved the highest PSNR/SSIM, while CT + mask was more medically meaningful because it combines anatomical context with lesion localization.

Condition	PSNR[5]	SSIM[6]
Mask	36.94	0.945
CT	37.01	0.963
CT + Mask	35.30	0.961

[5] Peak Signal to Noise Ratio
[6] Structure similarity

Discussion

Conclusion:
- Standard GAN outputs lacked meaningful PET structure.
- Mask and CT conditioning improved generation quality.
- CT + mask gave the most coherent PET-like outputs.

Future work:
- Test more stable GAN variants, such as WGAN-GP[7] or pix2pix-style cGAN, to improve training stability.
- Extend evaluation with more PET/CT cases and expert visual assessment to confirm medical relevance.
- Compare segmentation results when training with **real PET only** versus **real + synthetic PET** data.

[7] Wasserstein GAN

References

- Islam, J., & Zhang, Y. (2020). *GAN-based synthetic brain PET image generation.*
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al. (2014). *Generative adversarial nets.*

INSTITUT DE RECHERCHE EN INFORMATIQUE DE TOULOUSE 2 rue charles camichel , 31000 Toulouse

Project details